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Self-organization and symmetry breaking in intestinal organoid development

Denise Serra[1,2,6], Urs Mayr[1,2,6], Andrea Boni[1,5,6], Ilya Lukonin[1,2], Markus Rempfler[1], Ludivine Challet Meylan[1], Michael B. Stadler[1,3], Petr Strnad[1,5], Panagiotis Papasaikas[1,3], Dario Vischi[1], Annick Waldt[4], Guglielmo Roma[4] & Prisca Liberali[1,2]*

Intestinal organoids are complex three-dimensional structures that mimic the cell-type composition and tissue organization of the intestine by recapitulating the self-organizing ability of cell populations derived from a single intestinal stem cell. Crucial in this process is a first symmetry-breaking event, in which only a fraction of identical cells in a symmetrical sphere differentiate into Paneth cells, which generate the stem-cell niche and lead to asymmetric structures such as the crypts and villi. Here we combine single-cell quantitative genomic and imaging approaches to characterize the development of intestinal organoids from single cells. We show that their development follows a regeneration process that is driven by transient activation of the transcriptional regulator YAP1. Cell-to-cell variability in YAP1, emerging in symmetrical spheres, initiates Notch and DLL1 activation, and drives the symmetry-breaking event and formation of the first Paneth cell. Our findings reveal how single cells exposed to a uniform growth-promoting environment have the intrinsic ability to generate emergent, self-organized behaviour that results in the formation of complex multicellular asymmetric structures.

Genetically identical cells grown under the same conditions can display extensive variability in their potential to grow and differentiate[1–3]. This may be attributed to stochastic fluctuations in gene expression[4,5], or non-genetic variability emerging from collective cell behaviour[6–8]. The latter is generated by single cells that sense the 'population context', resulting in microenvironmental effects that can feed back on their physiological state and gene expression programs[9]. This enables single cells to break the symmetry of a population by changing their differentiation potential[10] with respect to other identical cells.

Organoids recapitulate the self-organizing potential of stem cells, creating three-dimensional structures in vitro. In particular, intestinal organoids recapitulate patterning processes and contain all cell types found in the adult intestine[11,12]. A characteristic of these intestinal organoids is that they develop from a single LGR5-positive ($LGR5^+$) stem cell[11]. Despite their extensive use, it is unclear how single intestinal stem cells give rise to cell populations with the capability of self-organization, and which transcriptional program it is that cells use. First, the stem cell generates a symmetrical sphere-like structure. Next, a secretory cell, named a Paneth cell, emerges and is believed to determine the future crypt site. Paneth cells create the niche environment and secrete WNT3A[13]. In vivo, these cells are not the only source of Wnt[14,15]. After this symmetry-breaking event, a gradient of WNT3A is formed around the Paneth cell, which induces the formation of a crypt[16]. The seemingly spontaneous emergence of a Paneth cell within a sphere—which is formed by genetically identical cells—represents the first and most crucial symmetry-breaking event in the formation of intestinal organoids, but how this occurs remains unknown.

Here we characterize the development of intestinal organoids using a combination of single-cell genomics and imaging approaches to show that generation of organoids is not limited to $LGR5^+$ cells, and organoid formation is a regenerative process that relies on transient YAP1 activation. Finally, we show that for effective organoid development, YAP1 needs to display transient cell-to-cell variability in localization, which in turn initiates a Notch and DLL1 lateral inhibition event that drives differentiation of Paneth cells and subsequent crypt formation.

Intestinal organoid development from single cells

Many intestinal cell types can de-differentiate in vivo during injury, and both $LGR5^+$ and $LGR5^-$ cells can generate organoids[17–19]. We characterized the growth of intestinal organoids from sorted single $LGR5^+$ and $LGR5^-$ cells derived from a LGR5::DTR-eGFP mouse[20] (Fig. 1a). Cells were seeded as multiple individual cells and cultured in ENR medium, containing WNT3A for the first three days only[13]. Organoids were fixed at different time points, stained with multiplexed immunofluorescence (4i)[21], imaged at high resolution and segmented using cellular computer vision algorithms (Fig. 1a, b, Extended Data Fig. 1a–c).

The efficiency of organoid formation was approximately 18%, for $LGR5^+$ and 7.5% for $LGR5^-$ (Fig. 1c). Both starting populations induce a similar, stereotyped pattern of organoid development: single cells at 24 h; small spheres with a lumen at 48 h; larger spheres of which most contain one Paneth cells at 72 h; asymmetric spheres with initial buds at 96 h; and organoids with crypts at 120 h (Fig. 1b). Organoids increase exponentially in size and cell number during the first 72 h (approximately 500 cells by 96 h, Fig. 1d, e). Notably, organoid eccentricity—a feature that measures object roundness—displayed bimodal distributions at 120 h (Extended Data Fig. 1d), which indicates that two morphologically distinct organoids developed (Fig. 1f). Markers of different intestinal cell types revealed the existence of spherical organoids that consist entirely of enterocytes, and are devoid of other cell types (Extended Data Fig. 1e, f). We refer to these organoids as 'enterocysts', as compared to budding organoids. Enterocysts appear after 60 h and, at 120 h, approximately 20% of multicellular structures are enterocysts, whereas around 80% are budding organoids (Fig. 1g). There is an increase in enterocysts from $LGR5^-$ cells, which suggests that some progenitors have a higher probability to become enterocysts (Fig. 1g). In the absence of Wnt, enterocysts die

[1]Friedrich Miescher Institute for Biomedical Research (FMI), Basel, Switzerland. [2]University of Basel, Basel, Switzerland. [3]Swiss Institute of Bioinformatics, Basel, Switzerland. [4]Novartis Institutes for BioMedical Research, Novartis Pharma AG, Basel, Switzerland. [5]Present address: Viventis Microscopy Sàrl, EPFL Innovation Park, Lausanne, Switzerland. [6]These authors contributed equally: Denise Serra, Urs Mayr, Andrea Boni. *e-mail: prisca.liberali@fmi.ch

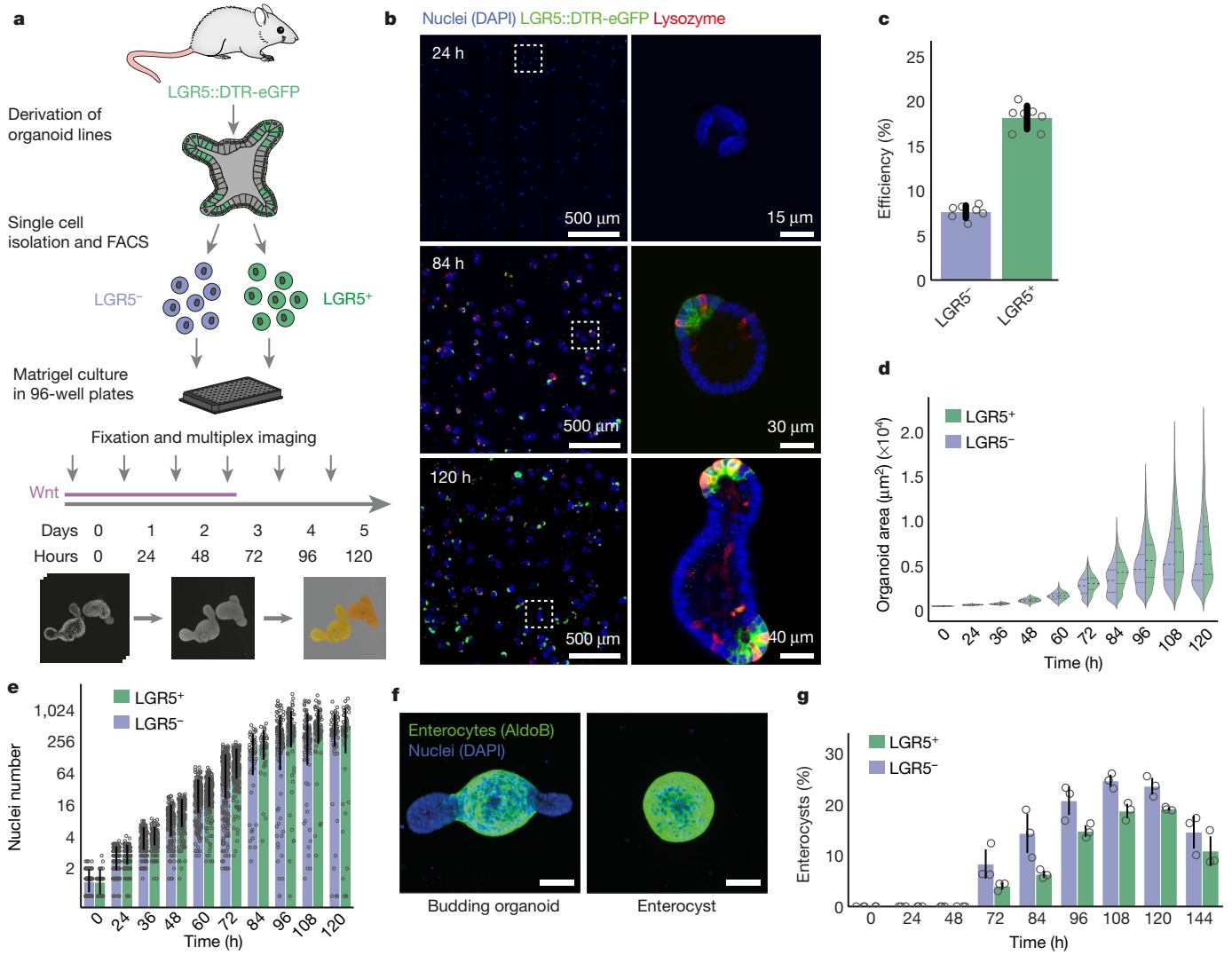


Fig. 1 | Intestinal organoids development from LGR5$^+$ and LGR5$^-$ single cells. a, Workflow of organoid-development time course from LGR5$^+$ and LGR5$^-$ single cells sorted by fluorescence-activated cell sorting (FACS) (LGR5::DTR-eGFP mice). **b**, Representative images of organoids (nuclei, DAPI), stem cells (LGR5::DTR-eGFP) and Paneth cells (lysozyme). Left, maximum intensity projections (MIP). Right, single-plane zoom-in. **c**, LGR5$^+$ higher efficiency of organoid formation ($n = 7$ replicates for each condition). $P = 7.6 \times 10^{-10}$, two-sided t-test. **d**, Organoid area (LGR5$^+$ $n = 9{,}798$ organoids; LGR5$^-$ $n = 13{,}623$ organoids). Violin plot lines denote quartile for each group. **e**, Nuclei number ($n = 2{,}829$ organoids). **f**, Representative images of budding organoids and enterocysts. Scale bar, 50 µm. **g**, Relative amount of enterocysts over time ($n = 3$ replicates for each condition). $P = 0.019$, two-sided t-test at 120 h. Data in **c**, **e**, **g** are mean ± s.d.

after 120 h (Fig. 1g). We propose that enterocysts develop owing to an absence of symmetry breaking and no Paneth cell differentiation (and thus an absence of endogenously produced Wnt). To test this hypothesis, we modulated the rate of Paneth cell differentiation, resulting in changes in enterocyst formation that show a strong correlation between the absence of Paneth cells and enterocyst formation (Extended Data Fig. 1g). Thus, the successful formation of budding organoids requires a symmetry-breaking event, which results in the differentiation into a Paneth cell in the presence of exogenous Wnt.

Trajectory of LGR5 dynamics and organoid development

We next quantified the time progression of organoid development to pinpoint when symmetry breaks (Fig. 2a, b). The subsampling of temporal progression with fixed organoid time courses prohibits the determination of the exact morphological and phenotypic stages at which enterocysts emerge. We therefore inferred continuous single-organoid trajectories of development using the imaging multidimensional feature space[22,23] (Extended Data Fig. 2a–d). Notably, organoids grown from LGR5$^+$ and LGR5$^-$ cells display identical patterns of growth (Fig. 2c, d, Extended Data Fig. 2e, f). This revealed a single 'pseudotime' trajectory up to approximately 0.3, after which it bifurcates into budding organoid and enterocyst branches (Fig. 2d, e). Both enterocysts and budding organoids are proliferative but only the budding organoids stay proliferative after removal of Wnt (Fig. 2e). Paneth cells appear only in budding organoids after pseudotime 0.3, and the timing is very similar for both starting populations (Fig. 2e, Extended Data Fig. 3). Mapping the cell number per organoid revealed that the bifurcation occurs when organoids have around 16–32 cells (Fig. 2f).

To map real time onto the trajectory, we performed time-lapse imaging from single cells using a custom-built light-sheet microscope suitable for organoids (Fig. 2g, Extended Data Fig. 4a–f). This revealed similar growth and proliferation dynamics for both budding organoids and enterocysts until 48 h (Fig. 2g, Extended Data Fig. 4g, Supplementary Videos 1–3). A sphere is created at the two-cell stage, with a subsequent fast rate of cell division of around 8 h. After 72 h, when exogenous Wnt is removed, budding organoids show localized cell proliferation, representing the site of crypt formation, whereas cells in enterocysts stop proliferating. Comparing organoid area over time with that of fixed organoids along the trajectory shows that pseudotime is slightly compressed in the beginning and stretched

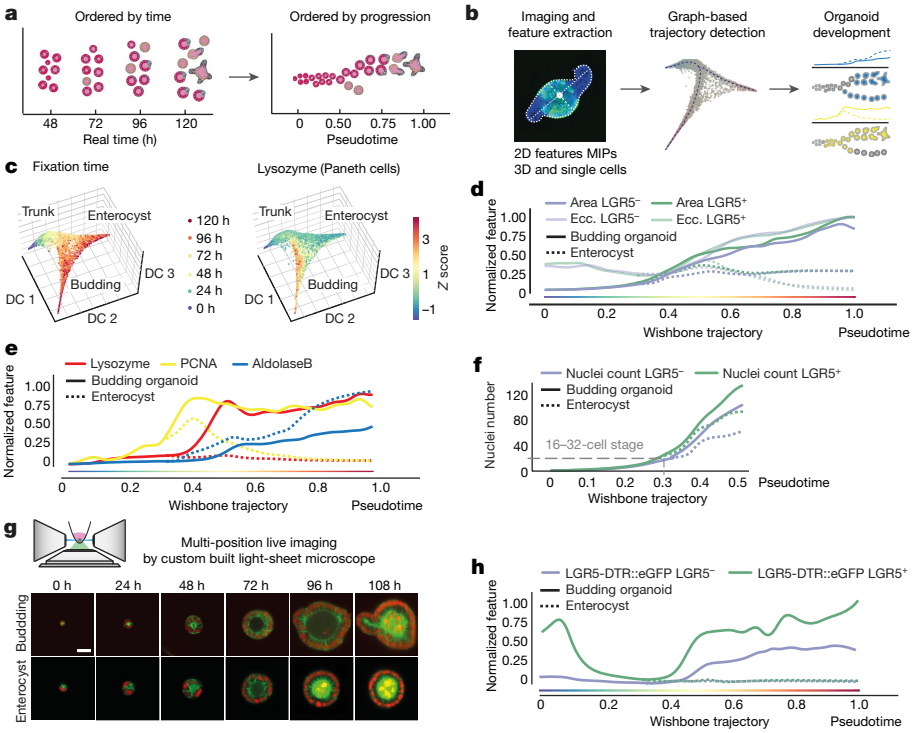


Fig. 2 | Trajectory of organoid development. a, **b**, Pseudotime ordering and trajectory inferences. **c**, Diffusion maps for pooled LGR5$^+$ and LGR5$^-$ organoids ($n = 23{,}421$). Colour-coding denotes fixation time and lysozyme intensity (Paneth cells). **d**, Morphological features on pseudotime of LGR5$^+$ ($n = 9{,}798$) and LGR5$^-$ ($n = 13{,}623$) organoids. Ecc., eccentricity. **e**, Cell types and proliferation markers on pseudotime ($n = 23{,}421$). **f**, Nuclei number on pseudotime for LGR5$^+$ ($n = 4{,}104$), LGR5$^-$ ($n = 5{,}593$) organoids. Dashed lines denote cell state around the branch point. **g**, Light-sheet imaging of single cells forming budding organoids or enterocysts (H2B–mCherry, red; MEM9–GFP, green). Single planes in the middle of organoids. Scale bar, 25 μm. **h**, LGR5::DTR-eGFP reporter signal on pseudotime of LGR5$^+$ ($n = 9{,}798$) and LGR5$^-$ ($n = 13{,}623$) organoids.

towards the end (Extended Data Fig. 4h). Mapping real time onto pseudotime further revealed that the bifurcation at approximately 0.3 pseudotime corresponds to around 45 h. Thus, a trajectory of organoid progression—inferred from thousands of fixed organoids obtained with a time-course experiment—provides an accurate representation of organoid development dynamic. Moreover, it allows us to directly compare experiments performed in a time course with live-cell imaging.

Following the expression of LGR5 along the trajectory led to an unexpected discovery. LGR5$^+$ cells strongly downregulate LGR5 expression during the first day and reacquire it only around 0.4 pseudotime (approximately 62 h) after the appearance of Paneth cells (Fig. 2h, Extended Data Fig. 5a, b). This loss of LGR5 expression also strongly correlates with the onset of fast proliferation and was confirmed with live-cell imaging of a single LGR5$^+$ cell (Extended Data Fig. 5c, d, Supplementary Video 4). LGR5 was also upregulated at 62 h in organoids from LGR5$^-$ cells, although slightly delayed and dimmer (Fig. 2h). This shows that LGR5$^+$ and LGR5$^-$ cells display similar patterns of organoid development through a transient proliferative sphere that consists of cells that do not express LGR5, after which LGR5 is expressed in stem cells in the newly formed crypts of budding organoids.

Transient YAP1 activation during organoid development

To determine the transcriptional programs modulated during the first days of organoid growth, RNA sequencing (RNA-seq) time-course analysis was performed and mapped onto pseudotime (Fig. 3a, Extended Data Fig. 6a, Supplementary Table 1). This shows that stem-cell markers are similarly downregulated at the transcriptional and protein level and, then, reacquired after Paneth cell differentiation (Fig. 3b, Extended Data Fig. 6b–d). Similar expression dynamics were also observed for all other cell-type-specific genes (Fig. 3b, Extended Data Fig. 6b–d). Notably, canonical Wnt targets are downregulated and non-canonical targets are upregulated in the first days (Extended Data Fig. 6e). To gain an overview of transcriptional changes, we then clustered the temporal expression patterns (Fig. 3c). This revealed three major clusters: first, a red cluster that is enriched in stem-cell markers, other cell-type-specific genes and genes involved in secretion and cell migration. There is a re-programming of cells into an undifferentiated state, which is followed by the re-establishment of LGR5$^+$ stem cells and acquisition of differentiation after Paneth cell formation (Extended Data Fig. 6f). Second, a blue cluster that is enriched in functions linked to mitochondria, actin cytoskeleton, cell cycle and extracellular matrix. These genes are probably required to supply the energetic demand of undergoing rapid cell divisions. Third, a green cluster that is enriched in functions related to metabolism and cell-type-related functions (Extended Data Fig. 6f).

To focus on the earliest phase of development, we quantified transcription-factor-binding motifs in the promoters of genes expressed at 24 h (Fig. 3d). The most important motifs were for FOSL1, TEAD1 and TEAD4, which all require YAP1 as transcriptional co-activator[24,25]. These transcription factors show an early increased expression, whereas the expression of *Yap1* mRNA stays constant (Fig. 3e, Extended Data Fig. 6g). YAP1 is a mechanosensing nuclear effector of the Hippo pathway and regulates organ growth, regeneration and tumorigenesis[26,27]. It is also an important effector of intestinal regeneration, in which it reprograms LGR5$^+$ cells into LGR5$^-$ cells, thus inhibiting Paneth cell differentiation[28,29]. When we correlated the expression levels of the early expressed genes (24 h versus 0 h) in organoid development with the expression levels of YAP1-dependent gene expression[28], we observed a good correlation ($r = 0.45$) (Fig. 3f). These early genes include YAP1 target genes[30] and fetal genes involved in regeneration[31]. Thus, this provides support for a model in which organoid development follows a regenerative response with a transient activation of YAP1 target genes in a LGR5$^-$ sphere.

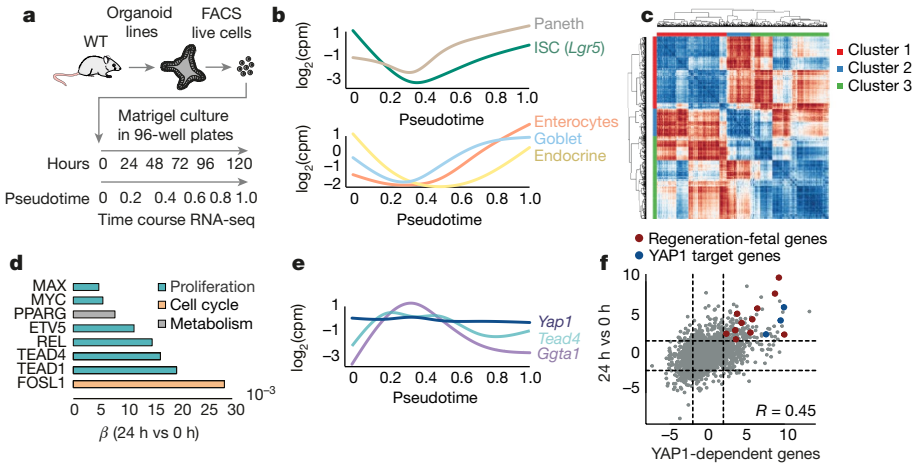


Fig. 3 | YAP1 target genes are transiently expressed during organoids development. a, Workflow for bulk RNA-seq time course. **b**, Expression profiles of cell-type markers mapped on pseudotime. Paneth cells (*Lyz1*), intestinal stem cells (ISCs; *Lgr5*), enterocytes (*Apoc3*), goblet cells (*Muc2*), endocrine cells (*Sst*). **c**, Hierarchical clustering of gene expression correlation profiles. **d**, Magnitude of contribution (β) of transcription factors to differential expression at 24 h versus 0 h. **e**, Gene expression profiles of *Yap1*, *Tead4* and *Ggta1* mapped on pseudotime. **f**, Pearson correlation between genes expressed at 24 h versus 0 h, and genes expressed in YAP1 overexpression versus knockdown. $n = 3$ organoid cultures from three independent mice.

Variable YAP1 activation determines symmetry breaking

YAP1 targets are transiently upregulated while *Yap1* mRNA is unchanged (Fig. 3e), which suggests post-transcriptional regulation. We therefore analysed YAP1 protein during organoid development. YAP1 protein abundance in organoids shows an initial increase until 48 h (Fig. 4a, b), whereas its subcellular localization starts to become variable between single cells after 24 h, increasing entropy in its ON and OFF state (Extended Data Fig. 7a, b). More precisely, in the four-cell stage, little cell-to-cell variability in the nuclear localization of YAP1 is observed whereas in the eight-cell and sixteen-cell stages—when cells are more crowded—YAP1 is translocated to the cytosol, and thus inactivated, in a subset of cells (Fig. 4a). After removal of Wnt, YAP1 protein levels strongly decreased (Fig. 4b), with YAP1 regionalizing only in the crypt area. YAP1 is absent in enterocysts. Removing Wnt earlier promotes relocalization of YAP1 to the cytosol, and decreased expression (Extended Data Fig. 7c).

When we generated organoids from single cells that, on average, have more YAP1 activity (all single cells extracted from 72-h-old organoids), we observed a strong increase in efficiency in organoid formation (Fig. 4c, Extended Data Fig. 7d). Inhibition of YAP1 and YAP1 knockout result in reduced organoid efficiency (Fig. 4d, e, Extended Data Fig. 7e). Conversely, ectopic activation of YAP1 by YAP1 overexpression, or by activation of its downstream effector EREG[28], increases the efficiency of organoid formation (Fig. 4f). All these phenotypes require the presence of exogenous Wnt (Extended Data Fig. 7f).

Homogeneous inhibition of YAP1 after 48 h in all cells reduces Paneth cell differentiation, and increases the number of enterocysts. Notably, this phenotype is similar to YAP1-overexpression organoids (in which YAP1 is homogeneously active in all cells), and neither form Paneth cells nor display symmetry breaking (Fig. 4g, h, Extended Data Fig. 7g, h). Instead, they either develop into enterocysts or remain as undifferentiated symmetrical spheres when YAP1 expression is high (Extended Data Fig. 7i). In addition, organoids that lack the LATS1 and LATS2 tumour suppressors[28] remain symmetrical and contain no Paneth cells (Extended Data Fig. 7g, h). We then added EREG, which led to the heterogeneous activation of YAP1, resulting in decreased enterocyst formation (Fig. 4g). From these findings, we conclude that during the first 72 h of organoid development, YAP1 is transiently active in every cell and induces cell proliferation, after which it becomes inactive in only few cells. Both homogeneous inactivation and overactivation of YAP1 in spheres abolish symmetry breaking and organoid formation, which suggests that it is not the absolute level of YAP1 but rather its varying activation state between single cells that drives symmetry breaking.

A YAP1–Notch switch drives symmetry breaking

To understand how YAP1 variability drives symmetry breaking, we performed single-cell (sc) RNA-seq at different time points (Fig. 5a, Extended Data Figs. 8a, b, 9a). At 72 h, the population of cells is homogeneous, with only a few stem cells and Paneth cells, and no enterocytes (Extended Data Fig. 9b). At 120 h, most cell types are represented[32,33] (Extended Data Fig. 9b, c). A subset of cells from organoids at 72 h shows high expression of YAP1 target genes (Fig. 5b, Supplementary Table 1), the levels of which correlate with the expression level of Notch ligands such as DLL1 (Fig. 5b, c, Extended Data Fig. 9d). This suggests that the variability in YAP1 activation might generate variability in expression of Notch ligands. Interestingly, Notch signalling is a regulator of the intestinal stem-cell niche, specifying distinct cell fates[34,35]. In vivo and in mature organoids, it is known that inhibition of Notch increases the differentiation of secretory cells[36]. Nuclear localization of YAP1 is variable between cells after the 4-cell stage and precedes the expression of DLL1 in few cells between the 8-cell and 16-cell stages. DLL1 expression occurs in cells with high levels of nuclear YAP1 (Fig. 5d–f, Extended Data Fig. 10a). This is consistent with findings in other tissues that *DLL1* is a YAP1 target gene[28,37]. Moreover, HES1 is expressed in single cells that neighbour DLL1$^+$ cells (Fig. 5d, Extended Data Fig. 10a). Between the 16-cell and the 32-cell stages, Paneth cells appear, which are all positive for DLL1 and start losing nuclear YAP1 (Fig. 5e, f, Extended Data Fig. 10a, b), as also seen in the loss of YAP1 target gene expression in Paneth cells from scRNA-seq (Fig. 5c). This indicates that variability in nuclear YAP1 is involved in the initiation of a Notch–DLL1 event, which, once established, maintains itself in the absence of YAP1[38,39]. We then homogeneously inactivated or activated YAP1, and in both conditions, no Notch–DLL1 event is observed (Extended Data Fig. 10c, d). By contrast, when we used EREG, Notch–DLL1 is activated (Extended Data Fig. 10c, d). Finally, to determine whether Notch–DLL1 activation is required for symmetry breaking, we used gamma-secretase inhibitors. All these inhibitors reduce symmetry breaking and Paneth cell differentiation, resulting in an increased fraction of enterocysts (Fig. 5g, Extended Data Fig. 10e), and strengthening the correlation between the absence of Paneth cells and enterocyst formation. Adding the Notch inhibitor *N*-[*N*-(3,5-difluorophenacetyl)-L-alanyl]-*S*-phenylglycine *t*-butyl ester (DAPT) after 120 h shows the expected increase in secretory cells (Extended Data Fig. 10f). Thus, although Notch lateral inhibition is required for

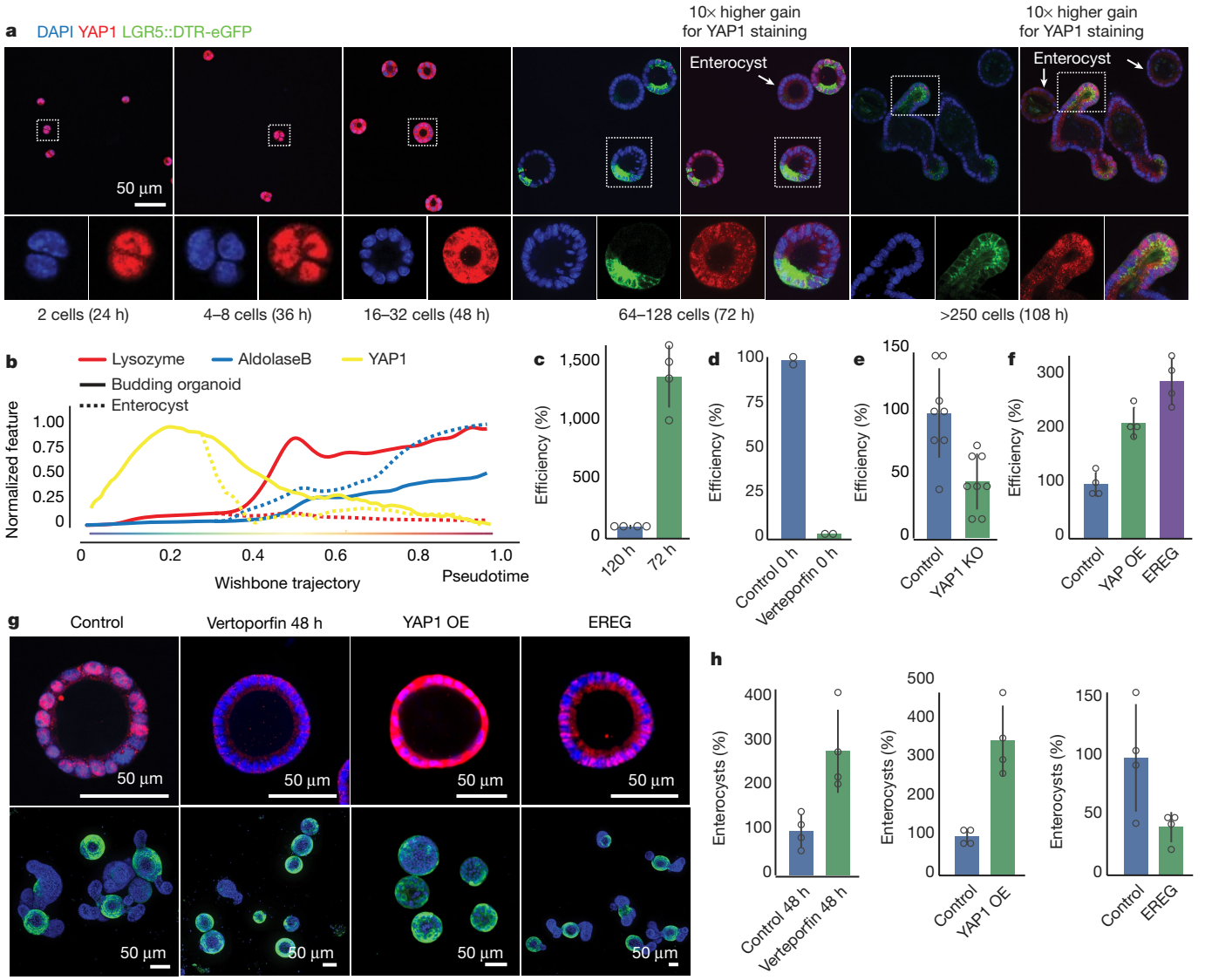


Fig. 4 | Transient YAP1 activation is necessary for symmetry breaking. **a**, YAP1 localization (LGR5::DTR-eGFP). **b**, YAP1 protein abundance along pseudotime ($n = 6{,}892$ organoids). **c**, Efficiency of organoids formation from cells isolated from organoids at 72 h and at 120 h (normalization: 120 h). Fixed at 72 h ($n = 4$ replicates). $P = 0.00007$, two-sided t-test. **d**, Efficiency of organoid formation after verteporfin treatment (normalization: control). Fixed at 48 h ($n = 2$ replicates). **e**, Efficiency of organoid formation from YAP1 knockout (KO) (normalization: control). Fixed: 72 h ($n = 8$ replicates). $P = 0.002$, two-sided t-test. **f**, Efficiency of organoid formation from YAP1 overexpression (OE) and EREG activation (normalization: control) ($n = 4$ replicates). $P = 0.001$ (YAP1 OE); $P = 0.0002$ (EREG), two-sided t-test. **g**, Images of organoids treated with verteporfin (at 48 h), YAP1-overexpression and EREG. Top, nuclei (blue) and YAP1 (red), fixed at 72 h. Bottom, nuclei (blue) and AldoB (green), fixed at 96 h (YAP1-overexpression) or 120 h (control, verteporfin, EREG). AldoB, aldolaseB. **h**, Quantification of enterocysts as in **g** (normalization: control) ($n = 4$ replicates). $P = 0.01$ (verteporfin); $P = 0.002$ (YAP1 OE); $P = 0.05$ (EREG), two-sided t-test. Data are mean ± s.d.

enterocyte differentiation in large organoids, it needs to be activated in small symmetrical spheres to drive symmetry breaking and the formation of the first secretory DLL1$^+$ cell, the Paneth cell. Together, this shows that variability in nuclear YAP1 initiates a Notch–DLL1 event. Cells with high levels of nuclear YAP1 become DLL1$^+$ cells and subsequently Paneth cells. Once having become a Paneth cell, the upstream regulation of YAP1 on DLL1 is then lost.

Discussion

Here we characterize the development of intestinal organoids from single cells, and show that it exploits the plasticity of the intestine and is driven by transient YAP1 activation, following a regeneration process. We then show that cell-to-cell variability in nuclear YAP1—emerging in symmetrical spheres—initiates a Notch–DLL1 lateral inhibition event between the 8- and 16-cell stage, which constitutes the first symmetry-breaking event in intestinal organoid growth and drives the formation of the first Paneth cell (Fig. 5h). It is yet not clear what drives the variability in YAP1 subcellular localization but it may be determined by a combinatorial effect of local variation in cell crowding caused by asynchronous cell divisions[40] and extracellular matrix density[41] (Extended Data Fig. 10g). Subsequently, variability in YAP1 results in variability in DLL1 activation. Although fluctuations in DLL1 ligand can be amplified by negative feedback[42], we show here that it does not arise from intrinsic stochasticity, but can be determined by cell-to-cell variability in the mechanosensor YAP1. Thus, Notch signalling has a dual role in organoid development—first in symmetry breaking and then, after homeostasis has been reached, in maintaining enterocyte differentiation[36].

Together, our findings underscore the notion that YAP1 acts as a sensor of tissue integrity. After tissue dissociation, YAP1 is activated to drive tissue repair, but once it is repaired, local cell crowding increases and induces heterogeneous activation of YAP1 in organoids and possibly also in vivo, driving the heterogeneous expression of DLL1 and

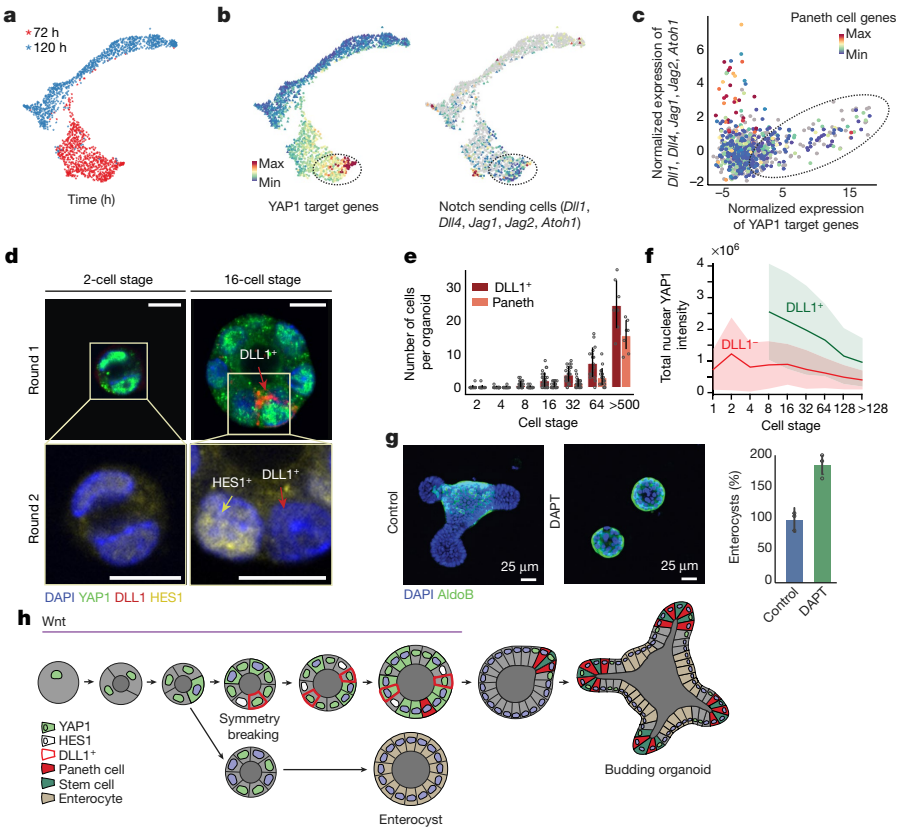


Fig. 5 | YAP1 cell-to-cell variability allows Notch–DLL1 activation and symmetry breaking. a, Griph (graph inference of population heterogeneity)-based visualization (72 h and 120 h) of single cells. **b**, Single-cell expression of YAP1 targets and Notch-sending cell marker genes. Dotted square denotes high YAP1 target-expressing cells. **c**, Spearman correlation between YAP1 target genes and expression of Notch-sending cell marker genes at 72 h ($n = 798$ cells). **d**, Multiplexed imaging of YAP1 variability and DLL1$^+$ cells. Top, nuclei (blue), YAP1 (green), DLL1 (red). Bottom, nuclei (blue) and HES1 (yellow). Arrows: red (DLL1$^+$ cells), yellow (HES1$^+$). Scale bar, 10 μm. **e**, Number of DLL1$^+$ and Paneth cells ($n = 170$ organoids). **f**, Nuclear YAP1 intensity in DLL1$^+$ (green) and DLL1$^-$ (red) cells ($n = 73{,}899$ cells). Shading denotes s.d. **g**, Left, images of organoids treated with DAPT. Fixed at 96 h. Right, quantification of fraction of enterocysts (normalization: control). Fixed at 96 h ($n = 3$ replicates). $P = 0.003$, two-sided t-test. **h**, Model of organoid development and symmetry breaking. Data in **e**, **g** are mean ± s.d.

formation of Paneth cells. Notably, cells that neighbour these Paneth cells show an increase in the canonical Wnt signalling response (TCF–GFP expression) (Extended Data Fig. 10h). Thus, Paneth cells might switch a general non-canonical/YAP1-dependent response in transient-amplifying-like cells to a locally induced canonical Wnt response in their neighbouring cells with the re-expression of LGR5 and the recreation of a stem-cell niche, bringing the system back to homeostasis. Indeed, YAP1 could be a general sensor for tissue injury and repair in many other tissues[43–45], by having a broad regulatory role at enhancers and distal regions of progenitor-specific genes[45,46]. This could shed light on the development of cancer[47], which may often reflect an inability to reacquire homeostasis upon tissue damage[28]. Finally, this work reveals how single cells have the intrinsic ability to generate emergent, self-organized behaviours that result in multicellular asymmetric structures.

Reporting summary

Further information on research design is available in the Nature Research Reporting Summary linked to this paper.

Data availability

RNA-seq data are available at the Gene Expression Omnibus (GEO) under accession codes GSE115955 (bulk) and GSE115956 (single cells).

Code availability

Code used for image analysis in this study was developed in the Liberali laboratory in Matlab and Python 3. For Python 3, multiple open source Python libraries for scientific computing and image analysis were used. The code for organoid 2D and 3D segmentation, feature extraction and organoid linkage is available under https://github.com/fmi-basel/glib-nature2018-materials. An R implementation of the Griph algorithm for scRNA-seq analysis is available as an R package at https://github.com/ppapasaikas/Griph.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, statements of data availability and associated accession codes are available at https://doi.org/10.1038/s41586-019-1146-y.

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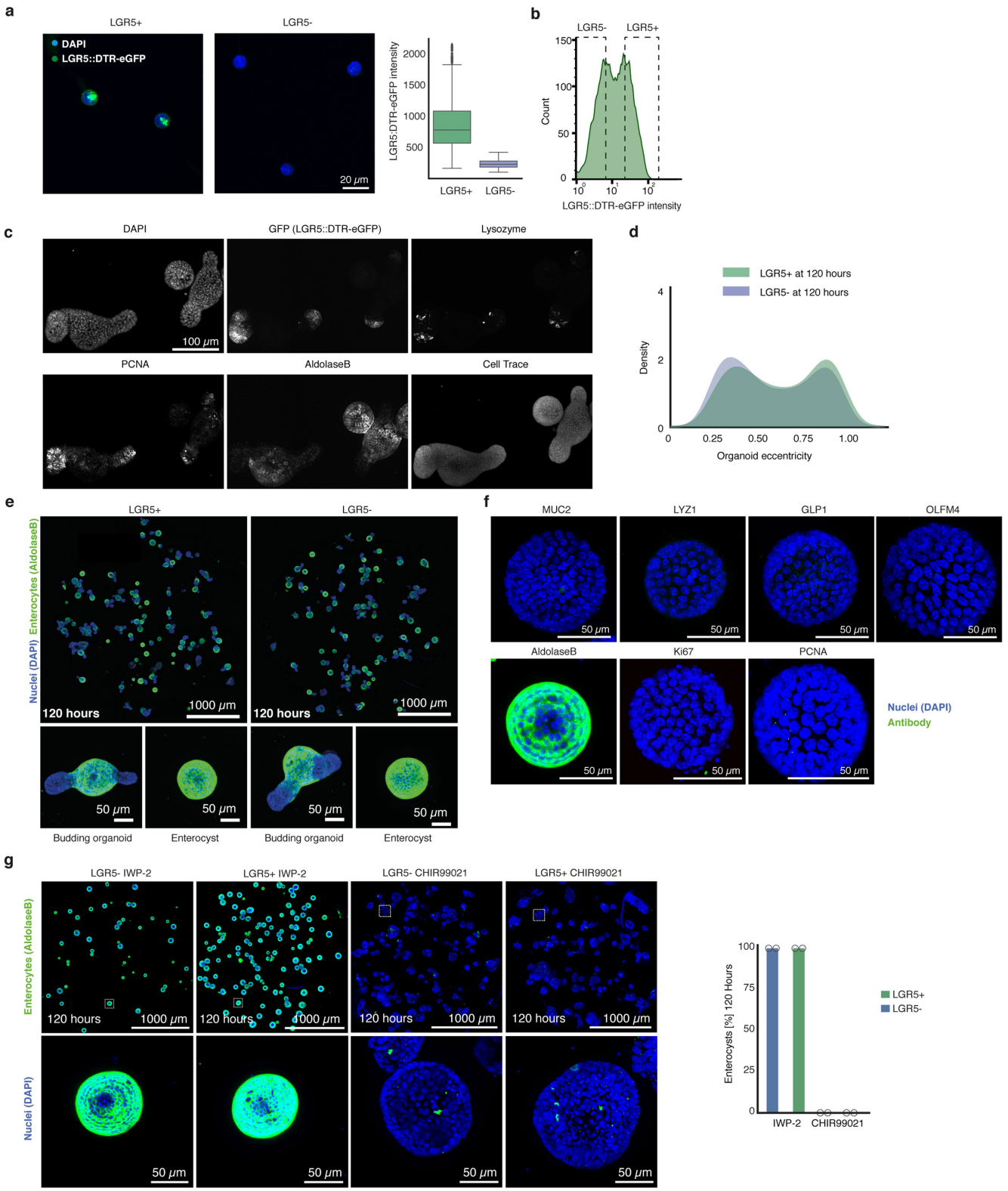
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Author contributions P.L. conceived and supervised the study, P.L., D.S., U.M. and A.B. designed the experiments, D.S., U.M., L.C.M. and A.B. performed the time courses, I.L. performed the first YAP1 experiments, U.M. and A.B. designed the image analysis, M.R. and D.V. developed 3D and single-cell imaging, U.M. developed the trajectory inference, A.B. and P.S. built the light-sheet, A.B. and U.M. performed the light-sheet experiments, A.B. performed light-sheet image analysis, D.S. performed all transcriptomics experiments, M.B.S. designed transcription factor motifs analysis, D.S. and M.B.S. performed the analysis of RNA-seq data, M.B.S. and P.P. developed Griph, A.W. and G.R. performed the scRNA-seq analysis, U.M. performed the multiplexing experiments, P.L., D.S., U.M. and A.B. wrote the paper.

Competing interests A.B. and P.S. are co-founders of Viventis Microscopy Sàrl that commercializes the light-sheet microscope used in this study.

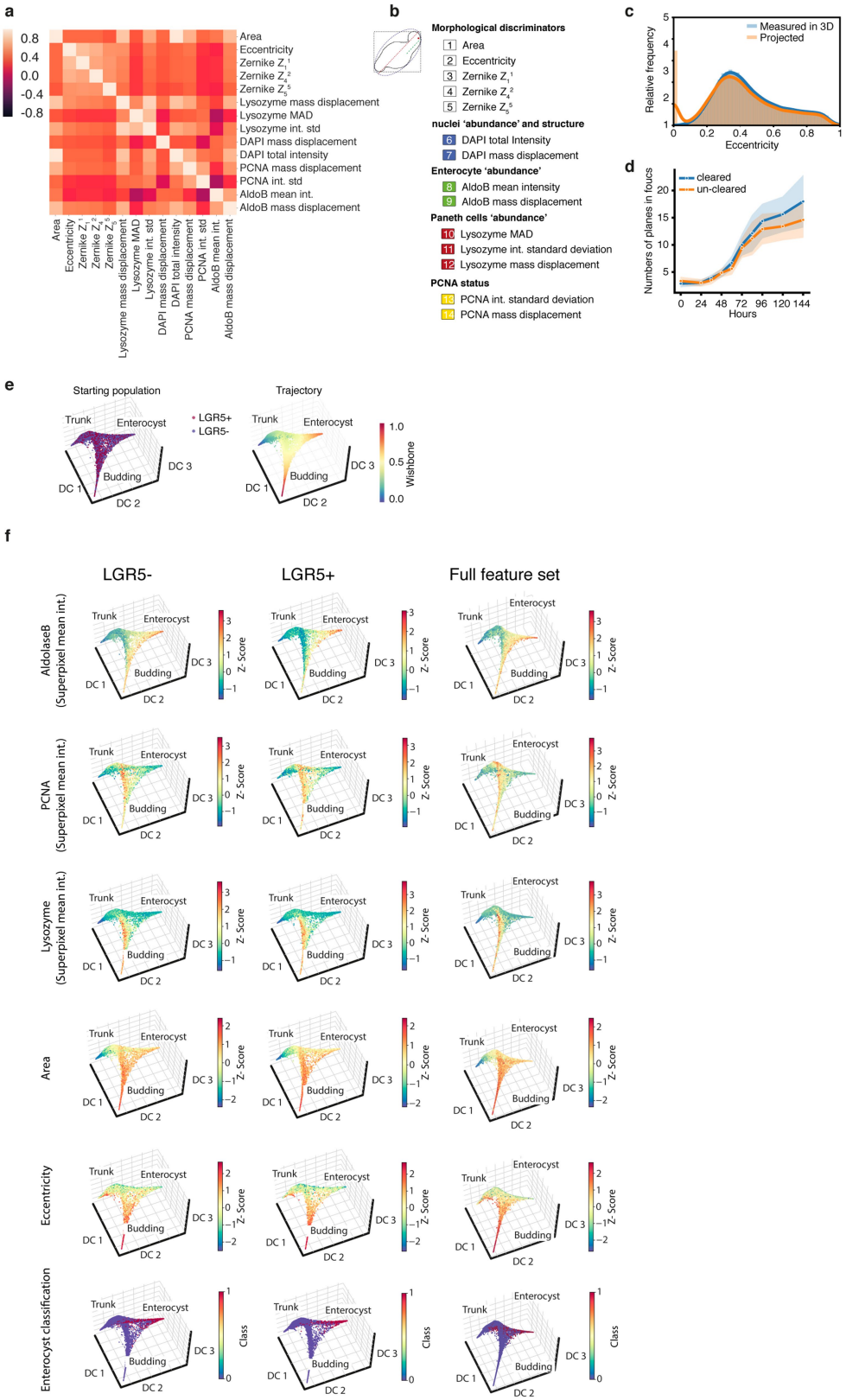
Additional information
Extended data is available for this paper at https://doi.org/10.1038/s41586-019-1146-y.
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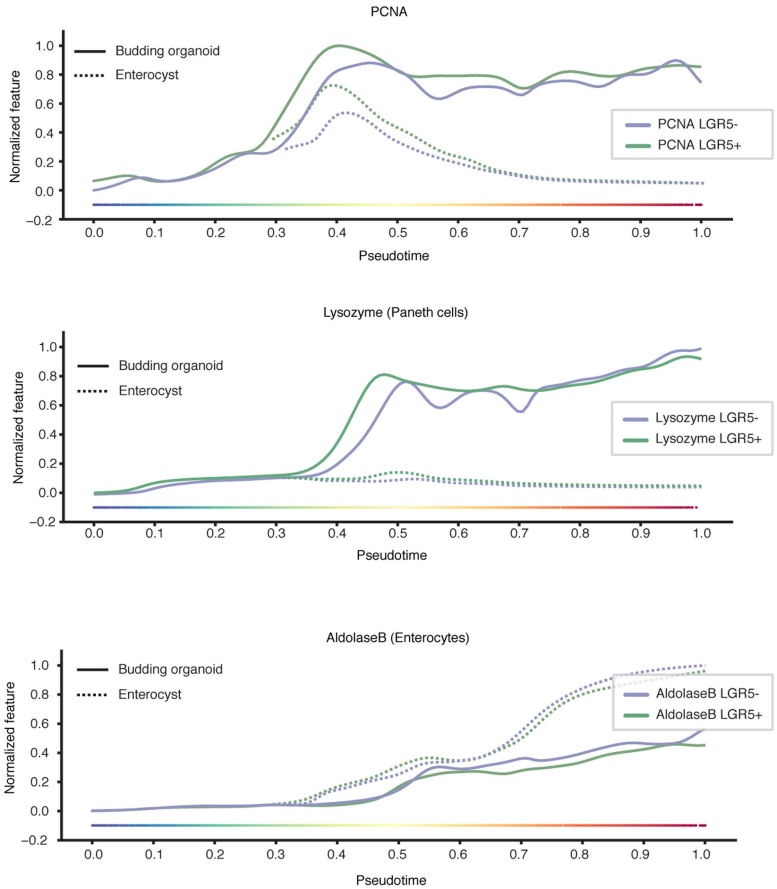


Extended Data Fig. 1 | See next page for caption.

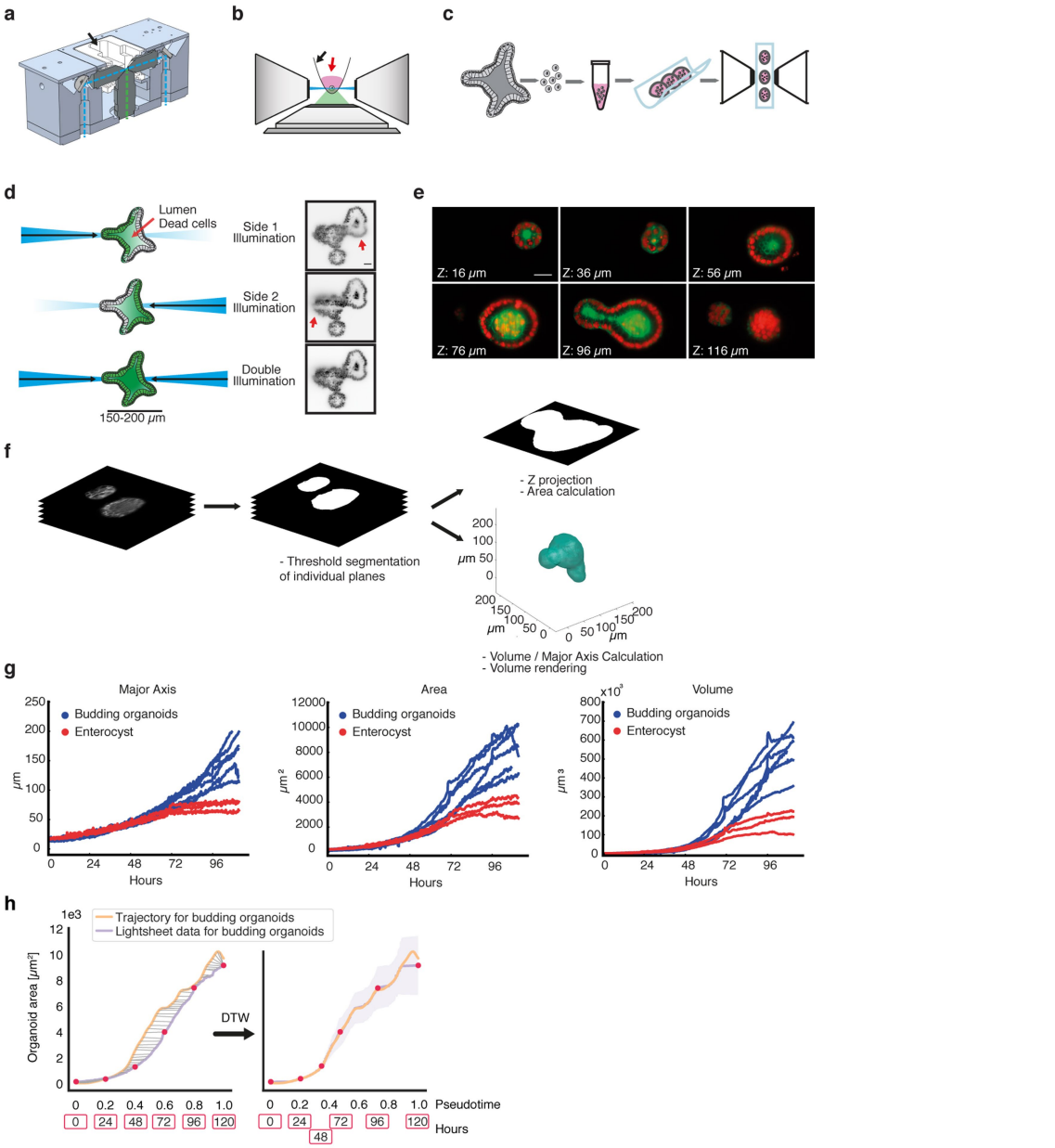
Extended Data Fig. 1 | Intestinal development of LGR5$^+$ and LGR5$^-$ organoids. a, LGR5$^+$ and LGR5$^-$ single cells sorted from LGR5::DTR-eGFP reporter mouse 3 h after FACS (left), with box plot showing LGR5::DTR-eGFP intensity quantification (right) ($n = 602$ organoids for LGR5$^+$; $n = 576$ organoids for LGR5$^-$). Box plot elements show quartiles, and whiskers denote 1.5× the interquartile range. **b**, FACS histogram of GFP signal of LGR5::DTR-eGFP single cells. Dashed boxes depict gating. **c**, Representative images of 4i imaging showing the same organoids stained with DAPI, GFP, lysozyme, proliferating cell nuclear antigen (PCNA), aldolaseB and Cell Trace. **d**, Distribution of eccentricity at 120 h of development for LGR5$^+$ and LGR5$^-$ starting populations (LGR5$^+$ $n = 463$ organoids; LGR5$^-$ $n = 711$ organoids). **e**, Representative images of LGR5$^+$ and LGR5$^-$ cells at 120 h of development. Bottom, enlargements depict budding organoids and enterocysts. **f**, Representative images of enterocysts stained with DAPI, different cell-type and proliferation markers. **g**, Left, representative images of organoids grown from LGR5$^+$ and LGR5$^-$ single cells in the presence of Wnt signalling inhibition (IWP-2, 2 μM) or activation (CHIR99021, 5 μM). Organoids are fixed at 120 h and stained for nuclear marker (DAPI) and enterocyte marker (aldolaseB). Top row shows complete well overview; bottom row shows magnified examples of single organoid. Right, quantification of enterocysts at 120 h of organoid development after Wnt signalling inhibition (IWP-2, 2 μM) or activation (CHIR99021, 2 μM) ($n = 2$ replicates). Data are mean values.



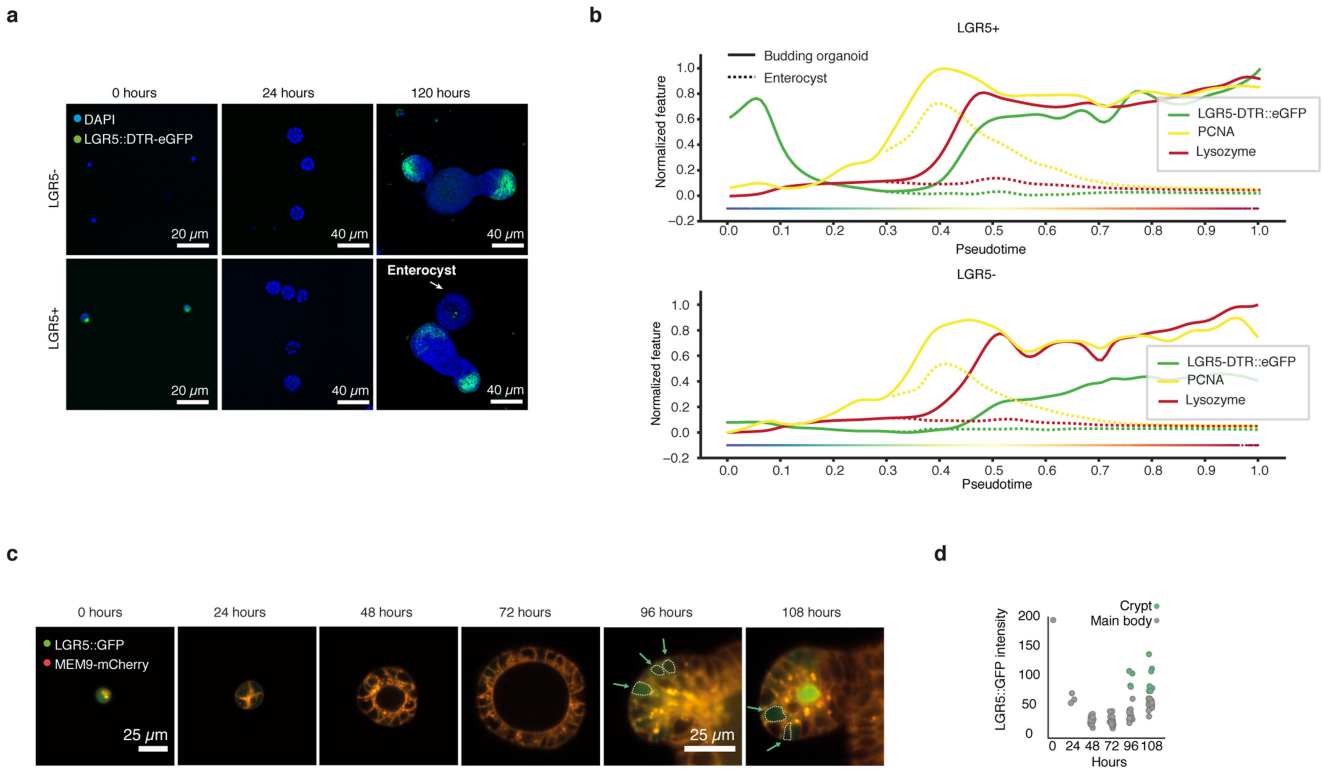
Extended Data Fig. 2 | Feature selection for organoid development. **a**, Pearson correlation heat map of the 14 selected features used for trajectory inference ($n = 23{,}421$ organoids). **b**, Fourteen selected features are grouped based on the underlying information. **c**, Comparison between eccentricities calculated on projected images and on 3D volumes ($n = 16{,}175$ organoids). **d**, Number of in-focus planes at different time points in cleared and uncleared organoids ($n = 2{,}310$ organoids). Planes are classified as in-focus with a logistic regression using the ratio of maximum in-plane intensity to maximum intensity in the whole stack as feature. Shading denotes s.d. **e**, Diffusion maps for pooled LGR5$^+$ and LGR5$^-$ organoids (14 selected features, $n = 23{,}421$ organoids). Colour-coding denotes starting population and pseudotime. **f**, Diffusion maps for LGR5$^-$ (first column, 14 selected features, $n = 13{,}623$) and LGR5$^+$ (second column, 14 selected features, $n = 9{,}798$) organoids, and for the pooled dataset with the full feature set (third column, 66 features, $n = 23{,}421$). Colour-coding denotes enterocytes (aldolaseB), PCNA, Paneth cells (lysozyme), organoid area, eccentricity and enterocyst classification (in which class 1 denotes enterocysts).



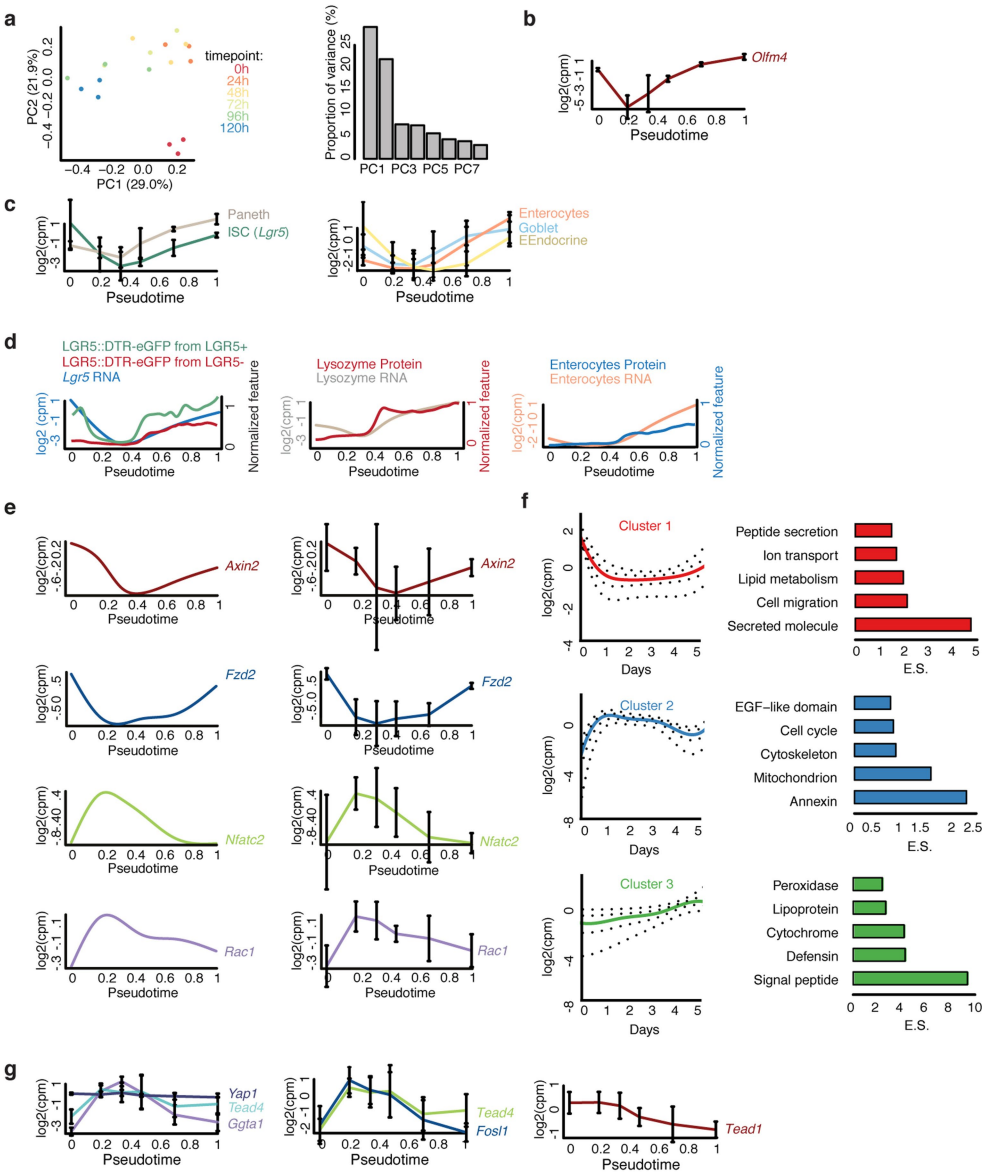
Extended Data Fig. 3 | Cell types and proliferation markers along organoid development trajectories. Proliferation (PCNA) and cell-type marker (lysozyme and aldolaseB) trends on pseudotime for LGR5^{+} ($n = 9{,}798$) and LGR5^{-} ($n = 13{,}623$) organoids.



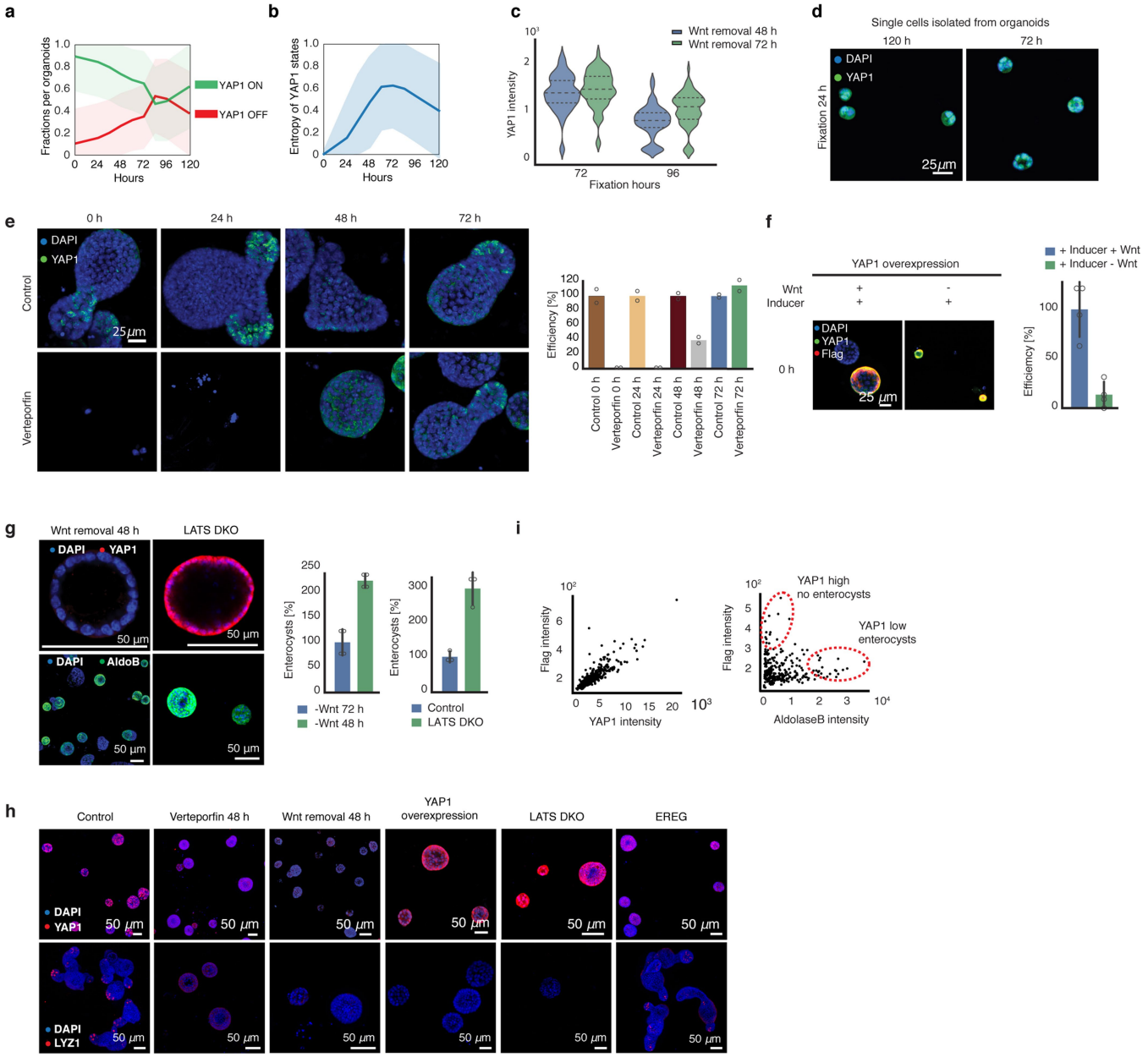
Extended Data Fig. 4 | Dual illumination inverted light-sheet imaging of organoids. a, Cross-sectional view of objective mounting block showing illumination paths (light blue dashed line) and imaging path (green dashed line). Illumination and imaging objective are solid mounted in the aluminium block and immersed in water contained in the reservoir (black arrow). **b**, Side view of microscope objectives arrangement: the sample embedded in Matrigel (red arrow) is held on the FEP membrane (black arrow) mounted in a sample holder and positioned between the two illumination objectives. **c**, Organoids handling procedure: grown organoids are dissociated and single cells are embedded in Matrigel. Several Matrigel drops can be transferred into the imaging chamber allowing imaging parallelization. **d**, Illustration of image improvement by double illumination. Scattering of single illumination beam by the organoid prevents complete and homogenous illumination (red arrows). This problem is minimized by double illumination. Scale bar, 25 µm. **e**, Representative images of an organoid expressing H2b–mCherry (red) and membrane-bound GFP (green). Slices every 20 µm across the organoid volume are shown. Scale bar, 25 µm. **f**, Workflow of light-sheet data analysis. **g**, Morphological features (major axis, area and volume) derived from light-sheet imaging. Budding organoids $n = 6$, enterocysts $n = 3$. **h**, Dynamic time-warp mapping of light-sheet data onto the trajectory. Budding organoid branch for mean area progression inferred from the trajectory (orange line, $n = 23{,}421$) and mean area progression extracted from time-lapse light-sheet imaging (violet line, $n = 6$) before (left) and after (right) morphing. Red dots indicate positions of real time (h) relative to pseudotime. Shading denotes s.d.



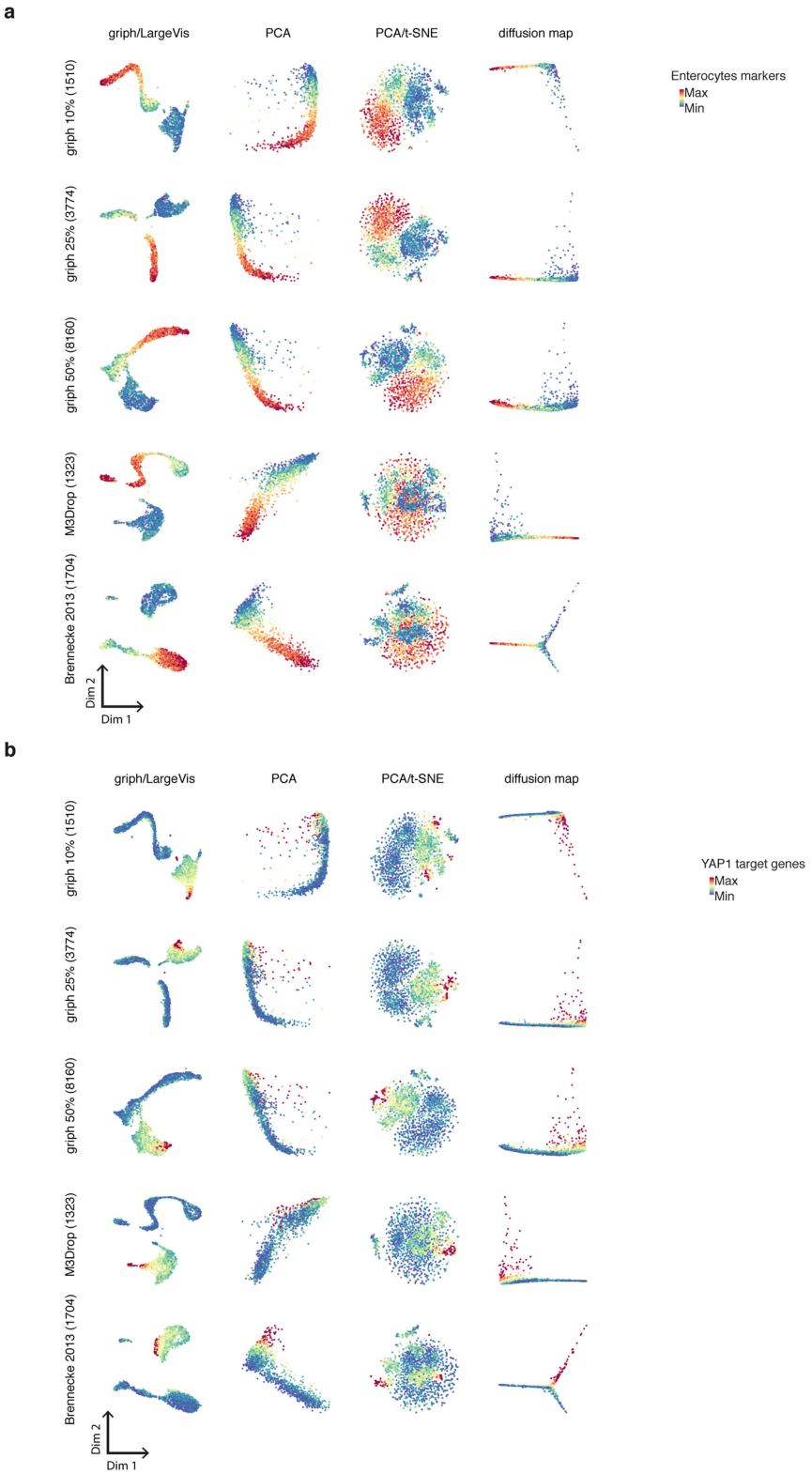
Extended Data Fig. 5 | LGR5 dynamics during organoid development. **a**, Representative images of organoids grown from LGR5$^+$ and LGR5$^-$ starting populations at 0, 24 and 120 h of development. **b**, Cell types and proliferation markers plotted on pseudotime for LGR5$^+$ (top, $n = 9{,}798$) and LGR5$^-$ (bottom, $n = 13{,}623$) starting populations. **c**, Light-sheet time-lapse imaging of organoid formation starting from a single LGR5$^+$ cell (GFP signal) expressing membrane-bound mCherry (MEM9–mCherry). Green arrows are pointing to cells re-acquiring GFP signal from LGR5 reporter in the organoid crypts. **d**, Quantification of GFP signal of LGR5 reporter from time lapse shown in **c**. Cells localized in the organoid crypt are plotted in green, cells localized outside the crypt and on the main body are plotted in grey ($n = 91$ cells).



Extended Data Fig. 6 | Bulk RNA-seq analysis. a, Principal component analysis (PCA) analysis of time-course (0–120 h) bulk RNA-seq shows no mouse or batch effect ($n = 3$ biological replicates). **b**, Unsmoothed gene expression profile mapped on pseudotime trajectory of stem cell marker *Olfm4*. **c**, Unsmoothed gene expression profiles mapped on pseudotime trajectory of cell-type-specific genes. Paneth cell: *Lyz1*; ISC: *Lgr5*; enterocyte: *Apoc3*; goblet cell: *Muc2*; enteroendocrine cell: *Sst*. **d**, Left, protein abundance and gene expression profile of ISC marker *Lgr5* mapped on pseudotime trajectory. Middle, protein abundance and gene expression profile of Paneth cell marker *Lyz1* mapped on pseudotime trajectory. Right, protein abundance of enterocyte marker aldolaseB and gene expression profile of enterocyte marker *Apoc3* mapped on pseudotime trajectory. **e**, Smoothed and unsmoothed gene expression profile of canonical Wnt target gene *Axin2*, *Fzd2* and non-canonical target genes *Nfatc2* and *Rac1*. **f**, Mean expression profile for each cluster and relative quantiles (0.05, 0.25, 0.50, 0.75 and 0.95). Annotation enrichments are shown on the right. E.S., enrichment score. **g**, Unsmoothed gene expression profile of *Yap1*, *Tead4* and *Ggta1* and top three transcription factors (*Tead1*, *Tead4* and *Fosl1*) contributing to differential gene expression between 24 h and 0 h. Data in **b**, **c**, **e**, **g** are mean and s.d. ($n = 3$ biological replicates).

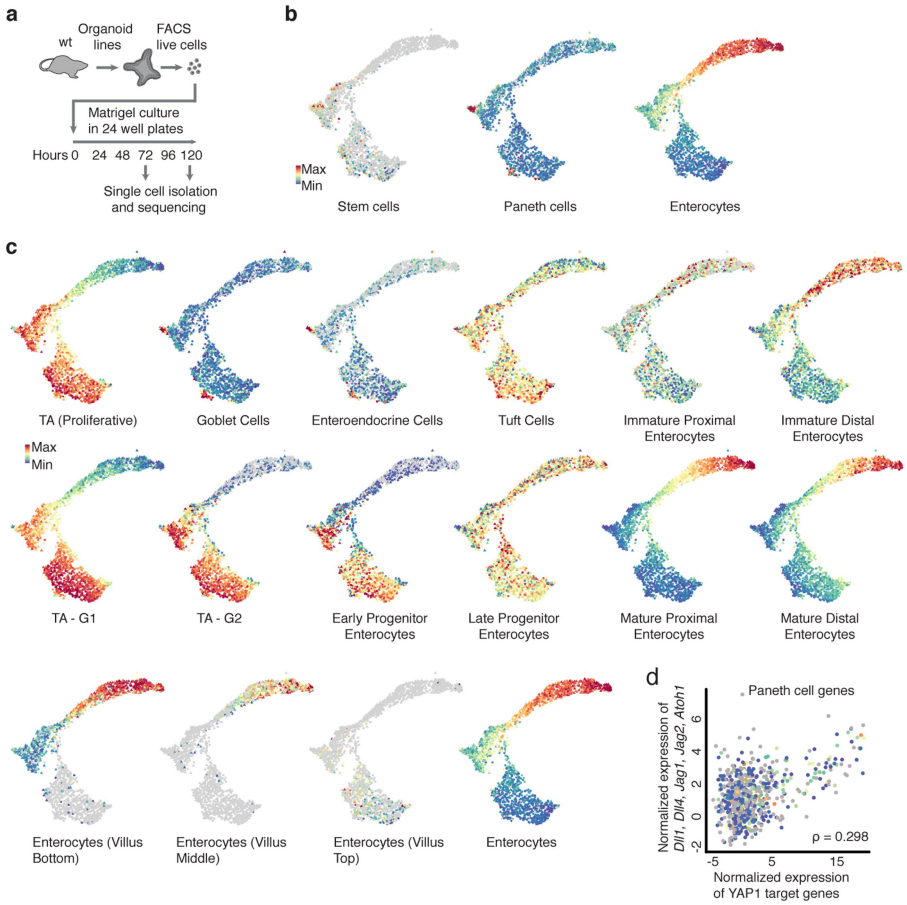


Extended Data Fig. 7 | YAP1 is necessary but not sufficient for organoid formation. a, Fraction of cells per organoid having YAP1 in the ON state (green, nuclear localization) or in the OFF state (red, cytoplasm localization) over time ($n = 1{,}074$ organoids). **b**, Entropy of YAP1 states (that is, variability in YAP1 activation in ON and OFF states) over time ($n = 1{,}074$ organoids). Shading denotes s.d. **c**, Quantification of YAP1 mean intensity after Wnt removal at 48 h or 72 h in organoids fixed at 72 h or 96 h ($n = 4$ replicates). Violin plot lines show quartiles. **d**, Images of organoids derived from single cells isolated from organoids at 72 h or at 120 h. **e**, Organoids treated at different time points with control or with verteporfin. Fixation at 96 h. Left, representative images; right, efficiency quantification (normalization: control, $n = 2$ replicates). Bar plots depict the mean. **f**, YAP1 overexpressing organoids with or without Wnt. Fixation at 96 h. Left, representative images; right, efficiency quantification (normalization: control, $n = 4$ replicates). $P = 0.001$, two-sided t-test. **g**, Left, images of organoids with Wnt removal at 48 h, or after double knockout of LATS1 and LATS2 (LATS DKO). Top, fixation at 72 h. Bottom, fixation at 96 h (Wnt removal at 48 h) or 120 h (LATS DKO). Right, quantification of enterocysts as retrieved in the left panel ($n = 4$ replicates for Wnt removal at 48 h; $n = 3$ replicates for LATS DKO). $P = 0.0001$ (Wnt removal); $P = 0.002$ (LATS DKO), two-sided t-test. **h**, Representative images of control, verteporfin treatment at 48 h, Wnt removal at 48 h, YAP1 overexpression, LATS DKO or treatment with EREG at 0 h. Top, fixation at 72 h; bottom, fixation at 96 h (Wnt removal at 48 h, YAP1 overexpression, LATS DKO) or 120 h (control, verteporfin 48 h, EREG). **i**, Scatter plot of Flag mean intensity signal and YAP1 mean intensity signal (left) and of Flag mean intensity and aldolaseB mean intensity (right) in YAP1-overexpressing organoids at 96 h. Data are mean ± s.d.

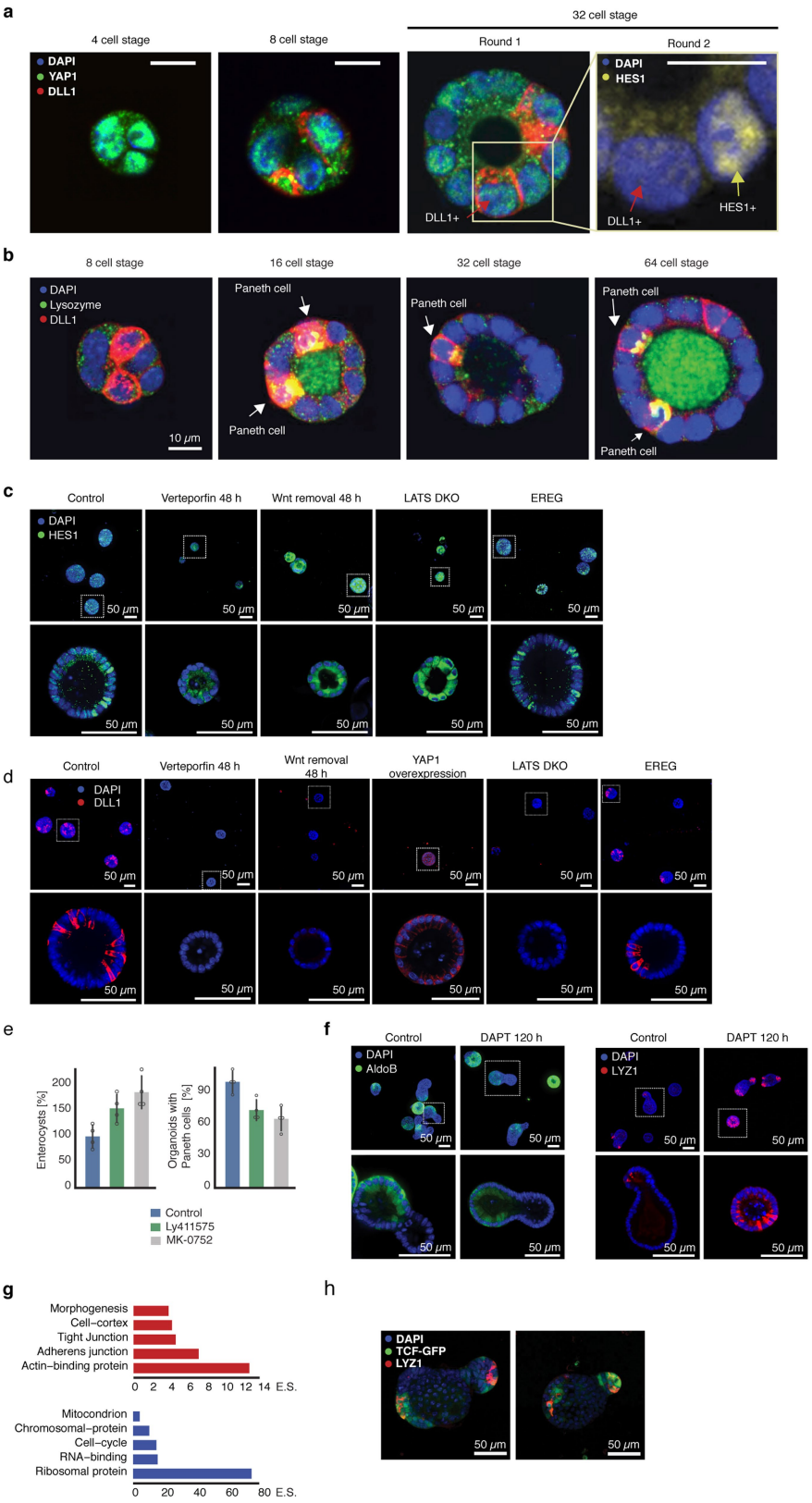


Extended Data Fig. 8 | Griph robustness analysis and comparison to other methods. Comparison of Griph lower dimensional embedding (Griph/LargeVis) to different embedding approaches (PCA, PCA combined with distributed stochastic neighbor embedding (*t*-SNE), diffusion maps) and analysis of method sensitivity for variable gene selection. Five different sets of variable genes have been selected (using Griph to retain 10%, 25% or 50% of genes per bin, by Michaelis–Menten fitting of the gene dropout rates as implemented in M3Drop, or by the previously described mean-variance fitting procedure[48]) and analysed. **a**, **b**, First and second dimension are shown and results are colour-coded for enterocyte marker genes (as in Extended Data Fig. 9b) (**a**) and YAP1 target genes (as in Fig. 5b) (**b**). $n = 1{,}863$ cells.

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Extended Data Fig. 9 | scRNA analysis. a, Experimental workflow for scRNA-seq. **b**, Griph-based visualization of single-cell degree of expression of marker genes for stem cells, Paneth cells and enterocytes. **c**, Griph-based visualization of single-cell degree of expression of marker genes for transient-amplifying cells, goblet cells, enteroendocrine cells, tuft cells, immature proximal enterocytes, immature distal enterocytes, transient-amplifying cells G1-phase, transient amplifying cells G2-phase, early progenitor enterocytes, late progenitor enterocytes, mature proximal enterocytes, mature distal enterocytes, enterocytes (villus bottom), enterocytes (villus middle), enterocytes (villus top) and enterocytes. **d**, Spearman correlation between expression of YAP1 target genes and expression of *Dll1*, *Dll4*, *Jag1*, *Jag2* and *Atoh1* in single cells at 120 h not expressing Paneth, goblet, enteroendocrine, enterocyte and stem-cell markers ($n = 696$ cells).



Extended Data Fig. 10 | See next page for caption.

Extended Data Fig. 10 | YAP1 cell-to-cell variability allows a Notch–DLL1 event and symmetry breaking. a, Images showing YAP1 variability and DLL1$^+$ cells. Magnified view denotes nuclear localization of HES1 in the same organoid. Arrows: red (DLL1$^+$ cells), yellow (HES1$^+$ cells). Scale bar, 10 μm. **b**, Images showing overlap of Paneth cells with DLL1$^+$ cells. **c**, MIP images (top) and magnified single plane images (bottom) of: control, verteporfin addition at 48 h, Wnt removal at 48 h, LATS double-knockout (DKO) and EREG treatment. Organoids are fixed at 72 h. **d**, MIP images (top) and magnified single plane images (bottom) of: control, verteporfin addition at 48 h, Wnt removal at 48 h, YAP1 overexpression, LATS DKO and EREG treatment. Organoids are fixed at 72 h. **e**, Fraction of enterocysts (left) and fraction of organoids with Paneth cells (right) for control and organoids treated with Ly411575 or MK-0752 at 0 h. Fixation at 120 h (normalization: control, $n = 4$ replicates). Ly411575 $P = 0.009$; MK-0752 $P = 0.003$, two-sided t-test. Data are mean ± s.d. **f**, MIP images (top) and magnified single plane images (bottom) of organoids treated with control or DAPT at 120 h and fixed at 144 h and stained for AldoB (left) or LYZ (right) **g**, Top, annotation enrichment of genes correlated with expression of YAP1 target genes in cells expressing YAP1 target genes and *Dll1*, *Dll4*, *Jag1*, *Jag2* and *Atoh1* genes. Bottom, annotation enrichment of genes anti-correlated with YAP1 target genes expression in cells expressing YAP1 target genes and *Dll1*, *Dll4*, *Jag1*, *Jag2* and *Atoh1* genes. E.S., enrichment score. **h**, MIP images of organoids at 120 h, showing the canonical Wnt signalling response (TCF–GFP) in cells neighbouring Paneth cells.

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Data collection	To collect imaging data, custom written segmentation code was used. The code was written in the Liberali lab and implemented with MATLAB 2016b (MathWorks) or Python 3.6 (relying on multiple open source Python libraries for scientific computing and image analysis). To collect imaging data Wako Software Suite version 1 was used. CellRanger suite 1.3.0 was used to collect sequencing data.
Data analysis	To analyze imaging data, Fiji (version 1.0) as well as custom written code in MATLAB 2016b (MathWorks) and Python 3.6 ((relying on multiple open source Python libraries for scientific computing and image analysis) was used. To analyze RNA-seq data, custom written R code (exact packages and parameters are described in the methods), STAR (version 2.5.2b), Griph 0.1.1, DAVID Bioinformatics Resources 6.8, JASPAR2016 Bioconductor package (version 1.6) was used.

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No sample size calculations were performed. In most cases, we assumed a minimum of around 100 organoids would be sufficient to recognize differences between control and perturbations based upon historical experiments in other contexts. In the time course experiments thousands of organoids were analyzed. Thousands of cells have been used for each time point of the single cell RNA sequencing
Data exclusions	Exclusion criteria for segmentation and RNA seq experiments were pre-defined. Parameters used to exclude datapoints were defined based on data distributions. Some organoids with segmentation artefact were excluded from the analysis with Z-score filtering. In the single cell RNA sequencing analysis high quality cells have been selected based on the maximum second derivative of the UMI count distribution, and genes with 0 counts or negligible variance (coefficient of variation, CV< 0.001) across all libraries were removed. In the bulk RNA sequencing analysis genes with less than two samples with at least one read per million reads were removed.
Replication	All experiments were replicated at least twice (if not indicated otherwise under Statistics and reproducibility) with similar finding. Multiplexing experiments and single cell RNAseq were only repeated once due to prohibitive costs.
Randomization	Samples were randomly assigned.
Blinding	The same investigators both performed experiments and data analyses therefore blinding was possible.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology
☐	☒ Animals and other organisms
☒	☐ Human research participants
☒	☐ Clinical data

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

anti-AldolaseB (Abcam ab75751, clone EPR3138Y, various lots for example GR121452-7, dilution 1:300),
anti-Yap (Cell signaling technology mAB #14074, clone D8H1X, lot #4, dilution 1:200),
anti-FLAG (Sigma-Aldrich F1804, clone M2, lot SLBT765, dilution 1:500),
anti-Lysozyme (DAKO A0099, clone NA (Polyclonal), lot 20040597, dilution 1:400),
anti-Dll1 (R&D Systems AF3970, clone NA (Polyclonal), various lots for example YXZ0114081, dilution 1:100),
anti-Hes1 (Cell signaling technology mAB #11988, clone D6P2U, lot #3, dilution 1:100),
anti-GFP (Abcam ab5450, clone NA (Polyclonal), lot GR3215617-1, dilution 1:500),
anti-PCNA (Cell signaling technology mAb #2586, clone PC10, lot #5, dilution 1:600),
anti-Muccin2 (Santa Cruz #SC-15334, clone NA (Polyclonal), lot K0315 no longer available, dilution 1:200),
anti-Glp1 (Abcam ab111125, clone EPR4042-407, lot GR74272, dilution 1:200),
anti-Olfm4 (Cell signaling technology mAB #39141, clone D6Y5A, lot #1, dilution 1:100),
anti-Ki67 (Abcam ab16667, clone SP6, lot GR216200-1, dilution 1:200),
Alexa Fluor 488 donkey anti rabbit IgG (Thermo fisher scientific A-21206, clone NA (Polyclonal), lot 1981155, dilution 1:500),
Alexa Fluor 568 donkey anti rabbit IgG (Thermo fisher scientific A10042, clone NA (Polyclonal), lot 1964370, dilution 1:500),
Alexa Fluor 488 donkey anti mouse IgG (Thermo fisher scientific A-21202, clone NA (Polyclonal), lot 1890861, dilution 1:500),
Alexa Fluor 568 donkey anti mouse IgG (Thermo fisher scientific A10037, clone NA (Polyclonal), lot 1752099, dilution 1:500),
Alexa Fluor 488 donkey anti goat IgG (Thermo fisher scientific A-11055, clone NA (Polyclonal), lot 1827671, dilution 1:500),
Alexa Fluor 568 donkey anti goat IgG (Thermo fisher scientific A-11057, clone NA (Polyclonal), lot 1711491, dilution 1:500),

Alexa Fluor 488 donkey anti sheep IgG (Thermo fisher scientific A-11015, clone NA (Polyclonal), lot 1900213, dilution 1:500),
Alexa Fluor 568 donkey anti sheep IgG (Thermo fisher scientific A-21099, clone NA (Polyclonal), lot 1878508, dilution 1:500),
Alexa Fluor 647 donkey anti sheep IgG (Thermo fisher scientific A-21448, clone NA (Polyclonal), lot 1914541, dilution 1:500).

Validation

Validation statements available from manufacturers:
AldolaseB (https://www.abcam.com/aldolase-b-antibody-epr3138y-ab75751.html),
anti-Yap (https://www.cellsignal.com/products/primary-antibodies/yap-d8h1x-xp-rabbit-mab/14074),
anti-FLAG (https://www.sigmaaldrich.com/catalog/product/sigma/f1804?lang=de&region=CH&gclid=EAIaIQobChMIyfGH3oXr3wIVC-aaCh3l3ALcEAAYASAAEgIdnfD_BwE),
anti-Lysozyme (https://www.agilent.com/en/product/immunohistochemistry/antibodies-controls/primary-antibodies/lysozyme-ec-3-2-1-17-(concentrate)-76124),
anti-Dll1 (https://www.rndsystems.com/products/mouse-rat-dll1-antibody_af3970,
additional validated in ref: "Distinct expression patterns of Notch ligands, Dll1 and Dll4, in normal and in inflamed mice intestine"),
anti-Hes1 (https://www.cellsignal.com/products/primary-antibodies/hes1-d6p2u-rabbit-mab/11988,
additional mouse intestinal crypt Immunohistochemistry shown in ref: "Paneth Cell Multipotency Induced by Notch Activation following Injury"),
anti-GFP (https://www.abcam.com/gfp-antibody-ab5450.html),
anti-PCNA (https://www.cellsignal.com/products/primary-antibodies/pcna-pc10-mouse-mab/2586),
anti-Muccin2 (Santa Cruz #SC-15334, No longer available from Santa Cruz),
anti-Glp1 (https://www.abcam.com/glp1-antibody-epr4042-407-ab111125.html),
anti-Olfm4 (https://www.cellsignal.com/products/primary-antibodies/olfm4-d6y5a-xp-rabbit-mab-mouse-specific/39141),
anti-Ki67 (https://www.abcam.com/ki67-antibody-sp6-ab16667.html),
Alexa Fluor 488 donkey anti rabbit IgG (https://www.thermofisher.com/antibody/product/Donkey-anti-Rabbit-IgG-H-L-Highly-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-21206),
Alexa Fluor 568 donkey anti rabbit IgG (https://www.thermofisher.com/antibody/product/Donkey-anti-Rabbit-IgG-H-L-Highly-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A10042),
Alexa Fluor 488 donkey anti mouse IgG (https://www.thermofisher.com/antibody/product/Donkey-anti-Mouse-IgG-H-L-Highly-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-21202)
Alexa Fluor 568 donkey anti mouse IgG (https://www.thermofisher.com/antibody/product/Donkey-anti-Mouse-IgG-H-L-Highly-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A10037),
Alexa Fluor 488 donkey anti goat IgG (https://www.thermofisher.com/antibody/product/Donkey-anti-Goat-IgG-H-L-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-11055),
Alexa Fluor 568 donkey anti goat IgG (https://www.thermofisher.com/antibody/product/Donkey-anti-Goat-IgG-H-L-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-11057),
Alexa Fluor 488 donkey anti sheep IgG (https://www.thermofisher.com/antibody/product/Donkey-anti-Sheep-IgG-H-L-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-11015),
Alexa Fluor 568 donkey anti sheep IgG (https://www.thermofisher.com/antibody/product/Donkey-anti-Sheep-IgG-H-L-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-21099)
Alexa Fluor 647 donkey anti sheep IgG (https://www.thermofisher.com/antibody/product/Donkey-anti-Sheep-IgG-H-L-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-21448).

Animals and other organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research

Laboratory animals

Male and female outbred mice between 7 and 15 weeks old were used for all experiments. Mouse lines used: C57BL/6 wild type (Charles River Laboratories), Lgr5–EGFP–Ires–CreERT2 (kind gift from Momo Bentires-Alj, University Hospital in Basel), Lgr5::DTR-EGFP (Genentech, de Sauvage laboratory), H2B-mCherry C57BL/6 x C3H F1 females heterozygous for H2B-mCherry (kind gift from T. Hiragi lab, EMBL), Lats1/; Lats2/ (LATS DKO, kind gift from Jeff Wrana, Department of Molecular Genetics, University of Toronto, Canada)48, Yap1tm1.1Dupa/J (Yap1flox)49 from The Jackson Laboratory.

Wild animals

The study did not involve wild animals

Field-collected samples

The study did not involve samples collected from the field

Ethics oversight

All animal experiments were approved by the Basel Cantonal Veterinary Authorities and conducted in accordance with the Guide for Care and Use of Laboratory Animals.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Flow Cytometry

Plots

Confirm that:

☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).

☒ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).

☒ All plots are contour plots with outliers or pseudocolor plots.

☒ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	Murine organoids were collected 5-7 days after passaging and digested with TripLE (Invitrogen) for 20 min at 37 °C. Dissociated cells were passed through a cell strainer with a pore size of 20 µm.
Instrument	BD Influx cell sorter (Becton Dickinson) and BD FACS Aria III
Software	For collection the BD FACS Sortware 1.2.0.142 (for BD Influx cell sorter) or BD FACS Diva 8.0.1 (for BD FACS Aria III). To visualize the gating in Extended Data Figure 1b, FLOWJO (FLOWJO, LLC) in version 10.2 was used.
Cell population abundance	After final gating, for both Lgr5- and Lgr5+, the average abundance was around 10%. Purity of Lgr5+ and Lgr5- cells was determined by fixation of plated cells and quantification of the Lgr5:DTR-EGFP signal 3h after sorting (see Extended data figure 1a)
Gating strategy	To remove debris, dead cells and cell doublets SSC-A/FSC-A gating, SSC-H/SSC-W gating and FSC-H/FSC-W gating were used. Lgr5 + and Lgr5- populations were separated based on the Lgr5:DTR-EGFP intensity (Extended data figure 1b).

☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.